A.6 ABLATION: THE IMPACT OF ARITHMETIC ACCURACY

An important question regarding the design of GSM-Symbolic is: when designing templates, "how should the numerical range of the variables be chosen?" and "how much of the performance drop can be attributed to arithmetic mistakes?".

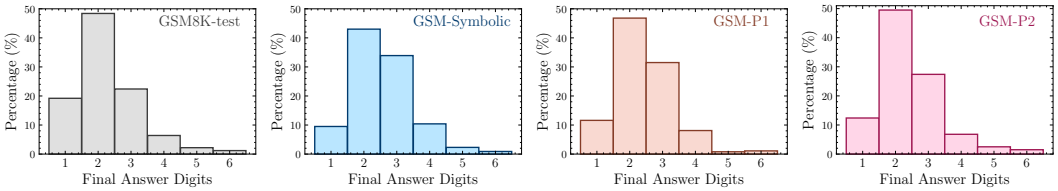


Figure 15: Statistics of the number of digits in the *final answer* for questions in GSM8K-test, GSM-Symbolic, GSM-P1, and GSM-P2. Compared to GSM8K-test, the GSM-Symbolic set shows a slight reduction in the number of questions with 1-digit or 2-digit final answers, and an increase in 3-digit answers. However, overall, the range of numbers does not increase significantly, with a significant majority of the final answers containing fewer than 5 digits.

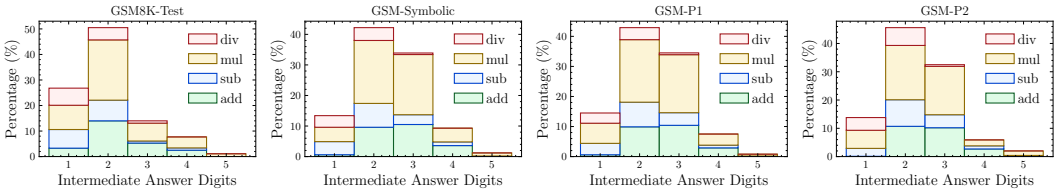


Figure 16: Statistics on the Number of Digits in *Intermediate answers*: Compared to GSM8K, GSM-Symbolic and its variants involve operations that result in 3-digit numbers. However, as shown in Fig. 17, modern LLMs are well capable of performing arithmetic within this range.

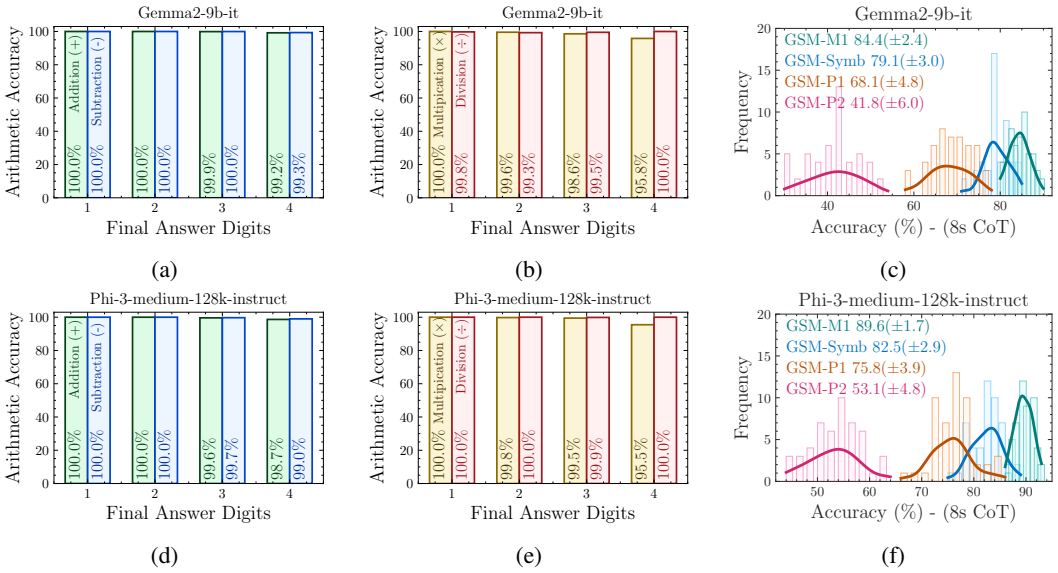


Figure 17: Arithmetic accuracy of Gemma2-9B and Phi3-Medium: **(a) and (d)** The addition accuracy of the models remains nearly perfect for calculations involving up to 4 digits. Note that, as shown in Fig. 15, the *majority of the final answers in GSM-Symbolic and its variants are below 5 digits.* **(b) and (e)** Although GSM-P1 and GSM-P2 have a similar digit-length distribution to GSM-Symbolic, as illustrated in Fig. 15, the performance drop is significant.

Overall, when creating the symbolic templates, we have chosen numerical ranges close to their original values in the GSM8K test set. The rationale behind this choice is that the arithmetic ability of the models is much less important than their logical reasoning capabilities. However, we face a challenge: we need to generate many instances per question. Given that not all values in the range satisfy the conditions and hence won't be selected, we need to slightly increase the numeric range of numbers to ensure enough instances are generated from each template.

However, we show that this adjustment does not push the numerical values into a range where models have low arithmetic accuracy. As observed in Fig. 15:

- Compared to GSM8K, GSM-Symbolic has more 3-digit final answers and fewer 1-digit and 2-digit answers. However, as shown in Fig. 17, modern LLMs such as Gemma2-9B have no difficulty with up to 3-digit addition and multiplication.
- Even though GSM-P1 and GSM-P2 have a very similar digit distribution to GSM-Symbolic and are mostly within the range that modern LLMs can perform accurate arithmetic (Fig. 17), the performance drop on these benchmarks is very significant (Fig. 17c).

Additionally, Fig. 16 provides statistics on the different operations involved in calculating *intermediate* answers across various datasets. Here, "intermediate" refers to the result of each intermediate operation throughout the solution process. For example, if a solution step involves $12 \times 6 = 72$, we categorize this as a two-digit multiplication. We observe that, overall, the distribution of the digit lengths in intermediate answers in Fig. 16, is similar to that in the final answers in Fig. 15.

Moreover, in Fig. 17, we demonstrate that modern Language Learning Models (LLMs) such as Gemma2-9B and Phi3-Medium are capable of nearly perfect addition and subtraction up to a length of 4, as well as achieving very high accuracy in multiplication and division. To this end, for each digit length of the answer, we ask the model in a zero-shot manner what the result of the operation would be. For example, for addition, we use the prompt *"What is x plus y?"* and for multiplication, we ask *"What is x times y?"*, and so on. To reduce the computational burden, we do not check arithmetic accuracy on every possible combination of numbers. However, we ensure that for every digit length and every operation, we have at least 1,000 test cases.

Finally, in Table 2 we report the arithmetic accuracy results of Gemma2-9b-it on a more realistic setup where calculations are extracted from the generated responses. To this end, we extracted parts of the generated responses where the equal sign (=) was generated along with numerical values, and then evaluated whether the left-hand side of the equation had the same numerical value as the right-hand side. As shown in the table, we observe very high arithmetic accuracy, similar to the previous results in this section.

Table 2: Ablation: Arithmetic accuracy of Gemma2-9b-it on different GSM benchmarks.

Benchmark	**1-digit**	**2-digits**	**3-digits**	**4-digits**	**5-digits**	**All**
GSM8K	99.8	99.1	99.3	96.8	95.9	98.9
GSM-Symbolic	99.7	99.2	97.6	97.2	99.1	98.3
GSM-P1	97.5	98.6	98.7	96.2	99.5	97.4
GSM-P2	96.2	98.6	97.5	96.4	99.2	97.1

Overall, we believe it is unlikely that arithmetic difficulty can account for the significant performance drop on benchmarks such as GSM-P2, which has a very similar distribution to GSM-Symbolic and falls well within the range of fewer than 5 digits that models like Gemma2-9B can handle accurately.

B Discussion on Statistical Analysis

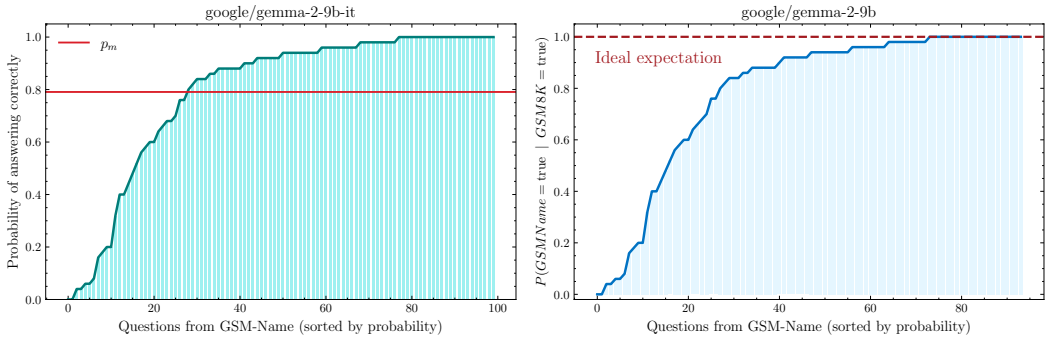
After the initial publication of our paper, we received various feedback regarding the statistical analysis of our results. Most notably, Ivanova et al. (2025) provide a detailed statistical analysis based on our results and argue that several of our observations are not statistically significant. First and foremost, we would like to thank the authors for taking the time to analyze our work. We firmly believe in the importance of the peer review process.

However, we note that we hold a fundamentally different philosophical view of reasoning than Ivanova et al. (2025). For instance, we do not believe the performance of an ideal reasoner—one that has truly understood a question—should follow a Bernoulli distribution. Rather, when focusing one single question, we argue that a degenerate distribution would be a more appropriate model. Although LLMs are probabilistic learners and do not follow such idealized behavior, this distinction is crucial to our analysis.

To be more specific, let us denote $P(o = \text{true})$ as the probability of a model answering question o from the GSM8K dataset correctly, and $P(x = \text{true})$ as the probability of correctly answering question x from GSM-Symbolic-Names, presented in Sec. 4.2, where the names have been changed but the variables remain the same—thus eliminating the influence of arithmetic mistakes.

We argue that treating $P(x = \text{true})$ as a Bernoulli trial with a model-specific success probability p_m is reasonable but not particularly useful for our purposes, as it ignores the model's response to the original question. Instead, we are interested in the conditional probability distributions $P(x = \text{true} \mid o = \text{true})$ and $P(x = \text{true} \mid o = \text{false})$, where the probability of answering x correctly is conditioned on whether the model correctly answers the original question o. Under this setup, our expectation is that for a "true reasoner," $P(x = \text{true} \mid o = \text{true})$ should follow a Bernoulli distribution with $p = 1$, and $P(x = \text{true} \mid o = \text{false})$ should follow a Bernoulli distribution with $p = 0$, since superficial changes (such as renaming variables) should not affect the outcome. Accordingly, the distribution of the total number of correct answers—conditioned on the model's responses to the original questions—would have an expected value of exactly $100 \cdot p_m$ and a standard deviation of 0.[5]. [6]

However, Figure 18 shows that LLMs such as gemma2-9b do not exhibit this behavior (other models follow a similar pattern).



(a) (b)

Figure 18: **(a)** The probability of answering each GSM-Name question correctly deviates from the uniform assumption of p_m. **(b)** The probability of answering GSM-Name questions correctly, conditioned on the model answering the original 100 GSM8K questions correctly, does not align with the expected behavior (i.e., Bernoulli with $p = 1$)

Overall, it is up to the reader to decide whether our stated assumptions are valid, or whether the statistical assumptions proposed by Ivanova et al. (2025) offer a better foundation for interpretation. Regardless, we acknowledge that we should have articulated our assumptions more clearly in the original version.

[5]One intuition behind our argument is the following: if a grade-school student can fully answer grade-school math questions but cannot answer high-school ones, then a test composed of 10 grade-school and 10 high-school questions would yield a performance of 50%. One could claim that the student's performance follows a Bernoulli trial with $p = 0.5$ (with expected score 10 and variance $np(1-p) = 5$), but this interpretation is not meaningful. A more realistic assumption is that the student always answers grade-school questions correctly and always fails high-school ones. In both cases, we do not expect any variance

[6]The arguments involving a degenerate distribution might seem extreme, as they assume the absence of non-reasoning factors (e.g., arithmetic mistakes, overlooking information during reasoning). In practice, this assumption does not always hold—humans also exhibit non-zero variance. However, we believe this variance is extremely small. It is worth noting that we are discussing simple grade-school questions involving basic arithmetic operations. For humans, especially when using a scratchpad, the likelihood of making mistakes on such calculations is very low.